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ORIGINAL ARTICLE

Comparative efficacy of isotonic saline (0.9% NaCl) and sea salt water mineral nasal spray on nasal mucosal eosinophil cytology in patients with moderate-to-severe persistent allergic rhinitis

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ABSTRACT

Background and aim: Nasal irrigation is a foundational adjuvant therapy for allergic rhinitis (AR). While 0.9% NaCl (isotonic saline) is conventional, sea salt water (SSW) mineral sprays enriched with complex trace elements are hypothesized to provide superior anti-inflammatory modulation. This study compared the efficacy of 0.9% NaCl and SSW mineral nasal spray on nasal mucosal eosinophil counts and clinical symptoms in patients with moderate-to-severe persistent AR.
Methods: In this prospective cohort study, 30 patients with moderate-to-severe persistent AR were allocated into two groups (n=15 each): one group received 0.9% NaCl irrigation (20 mL twice daily), and the other received SSW mineral spray (0.15 mL twice daily). Clinical symptoms were evaluated using the Visual Analog Scale (VAS) at baseline, week 2, and week 4. Objective nasal mucosal eosinophil counts were assessed via nasal scraping and Papanicolaou staining at baseline and week 4.
Results: Both groups exhibited significant reductions in nasal eosinophil counts and VAS scores by week 4 ($p<0.001$). Notably, the SSW group demonstrated significantly accelerated clinical improvement in rhinorrhea, sneezing, and ocular symptoms at week 2 compared to the 0.9% NaCl group ($p<0.001$). By week 4, no significant differences were observed between the two groups in the magnitude of eosinophil reduction ($p=0.423$) or final total VAS scores.

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Conclusion: Both 0.9% NaCl and SSW mineral nasal spray are equally effective for sustained symptomatic and inflammatory control in persistent AR. However, SSW mineral spray offers a distinct clinical advantage through accelerated symptomatic relief during the early phase of treatment.
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Key words: rhinitis allergic, nasal lavage, sodium chloride, seawater, eosinophils, nasal mucosa

Introduction

Allergic rhinitis (AR) is an IgE-mediated inflammatory disorder of the nasal mucosa triggered by allergen exposure, characterized by clinical symptoms such as rhinorrhea, nasal obstruction, paroxysmal sneezing, and nasal pruritus. Globally, AR represents a significant public health challenge with a steadily increasing prevalence, currently affecting 10–30% of the adult population worldwide (1,2). According to the AR and its Impact on Asthma (ARIA) classification, the moderate-to-severe persistent grade warrants specific clinical focus due to its extensive negative impact on quality of life, and work productivity (3). The pathogenesis of AR is increasingly exacerbated by the conditions of modern technological civilization; continuous exposure to environmental pollutants, transport and industrial emissions, household allergens, chemical irritants, and lifestyle factors such as tobacco smoke all contribute significantly to the initiation of an IgE-mediated inflammatory response. In Indonesia, the prevalence of this condition is reported to be between 1.5% and 12%, with a rising trend observed among the productive-age population in urban areas (4).

The pathogenesis of AR involves the infiltration of inflammatory cells into the nasal mucosa in response to type I hypersensitivity reactions. Eosinophils serve as the primary effectors in this process; their degranulation releases various cytotoxic proteins and pro-inflammatory mediators that exacerbate mucosal epithelial damage (5). Cytological examination of nasal mucosal eosinophils has been recognized as a sensitive objective indicator for assessing local inflammatory activity and correlates directly with the clinical severity experienced by patients (6). Consequently, the reduction of eosinophil counts in nasal secretions and the mucosa is established as a critical parameter for therapeutic success in the long-term management of AR.

Current management strategies for AR combine allergen avoidance, pharmacotherapy, and nasal irrigation as an adjuvant therapy. Nasal irrigation using 0.9% NaCl (isotonic saline) has long been widely utilized due to its efficacy in clearing allergens and improving mucociliary clearance (7). However, recent innovations in topical therapy have introduced the use of sea salt water (SSW) mineral nasal sprays, which feature a complex mineral composition including magnesium, calcium, and potassium. The magnesium content in sea water is hypothesized to exert a more potent anti-inflammatory effect by inhibiting the degranulation of mast cells and basal epithelial cells, theoretically providing superior efficacy compared to standard saline solutions (8).

Despite the common clinical use of both irrigation modalities, comparative studies specifically evaluating their impact on objective eosinophil cytological parameters in moderate-to-severe persistent AR patients remain scarce, particularly in Indonesia. Most previous research has focused predominantly on subjective symptom assessments via the Visual Analog Scale (VAS) without incorporating objective cellular inflammatory markers. Addressing this research gap, the present study aims to compare the effectiveness of 0.9% NaCl and SSW mineral nasal spray regarding changes in nasal mucosal eosinophil counts. The findings of this study are expected to provide robust clinical evidence for selecting the optimal nasal irrigation modality to reduce morbidity in patients with severe AR.

Materials and methods

Study design and setting

This study was observational research with a prospective cohort design, evaluating pre- and post-intervention parameters in a parallel manner. All research procedures were conducted at Dr. Wahidin Sudirohusodo Hospital, Makassar, Indonesia and the network hospitals from January to February 2025.

Participant selection and clinical criteria

Subjects were selected using a purposive sampling technique (9–11) from the patient population attending the Otorhinolaryngology-Head and Neck Surgery outpatient clinic. Inclusion criteria required subjects to be diagnosed with moderate-to-severe persistent AR based on ARIA criteria (12), supported by positive skin prick test (SPT) results (grade +3 or +4) for environmental allergens, and a total VAS score exceeding 5. Conversely, the study excluded individuals with a history of acute respiratory tract infections within the last 30 days, significant nasal anatomical abnormalities such as severe septal deviation or nasal polyps, patients with non-allergic rhinitis with eosinophilia syndrome (NARES; inherently ruled out by the mandatory positive SPT requirement), and the use of rhinitis medications (corticosteroids, antihistamines, or antibiotics) within four weeks prior to enrollment."

Intervention procedure and treatment protocol

A total of 30 eligible subjects were parallelly allocated into two treatment groups, with 15 subjects in each cohort. The first group was instructed to perform nasal irrigation using a 0.9% NaCl (isotonic saline) solution at a volume of approximately 20 ml per nostril. The second group received SSW mineral nasal spray at a dosage of approximately 0.15 ml per spray. Both procedures were performed routinely twice daily, in the morning and evening, for four weeks. To ensure optimal fluid distribution within the nasal cavity, subjects were trained in the nasal irrigation technique, which involved a 45-degree head tilt while breathing through the mouth.

Evaluation instruments and data measurement

Clinical efficacy was assessed through two primary parameters measured at baseline, week 2, and week 4. Subjective parameters were measured using a 100-mm VAS to evaluate the severity of rhinorrhea, nasal obstruction, sneezing, and pruritus.

Objective data were collected via cytological sampling through atraumatic scrapings of the inferior turbinate surface without anesthesia utilizing a sterile curette. To meticulously preserve cellular morphology and prevent air-drying artifacts, the collected material was immediately smeared onto glass slides and wet-fixed in 95% ethyl alcohol at the point of care within seconds of collection. Following immediate fixation, the specimens were secured in closed slide mailers and transported at room temperature to the laboratory within a maximum timeframe of 2 hours post-collection.

Upon arrival, staining was executed utilizing the progressive Papanicolaou technique. The fixed smears were hydrated through descending grades of ethanol to distilled water, followed by precise nuclear staining with Harris hematoxylin for 3–5 minutes. After a thorough tap water rinse to promote blueing, the slides were dehydrated through ascending ethanol grades. Cytoplasmic staining was subsequently performed utilizing Orange G-6 (OG-6) and Eosin-Azure 50 (EA-50) for 2 and 3 minutes, respectively. Finally, the slides were cleared in xylene and mounted with a synthetic resinous mounting medium.

Under light microscopy, this progressive methodology yielded distinct cytoplasmic transparency and sharp nuclear detail, allowing for the definitive identification of eosinophils—characterized by their classical bilobed nuclei and vibrant orange-red to pinkish specific granules (Figure 1). Eosinophil cell counting was performed by a competent clinician at 400x magnification across ten high-power fields (HPF), which

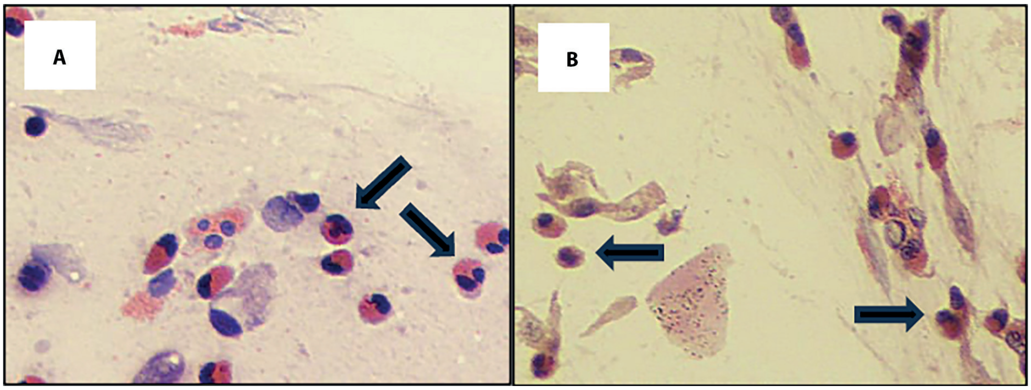

Figure 1. Cytological examination of nasal mucosal scrapings utilizing Papanicolaou staining (400x magnification). A) Multiple eosinophils (arrows) are prominently visualized, characterized by their classic dark purple bilobed nuclei and cytoplasm packed with vibrant orange-red eosinophilic granules. B) Scattered eosinophils (arrows) observed amidst the epithelial background, demonstrating marked eosinophilic infiltration. This distinct cytological profile is highly indicative of an allergic inflammatory response, characteristic of allergic rhinitis.

were then classified based on the Budiman score criteria (ranging from 0 to +4) (13,14).

Research ethics

Ethical approval was obtained from the Institutional Review Board of the Faculty of Medicine, Hasanuddin University (No. 428/UN4.6.4.5.31/PP36/2025) on June 23, 2025. Prior to data collection, all participants were provided with a comprehensive explanation regarding the objectives and risks of the study and were required to provide written informed consent.

Statistical analysis

All collected data were tabulated and processed using SPSS version 25.0 statistical software. Univariate analysis was performed to describe the demographic characteristics of the participants, while the Shapiro-Wilk normality test was used to determine the distribution of the data. Intra-group changes in clinical parameters were analyzed using the Wilcoxon or Friedman tests. To compare the effectiveness between treatment groups, the Mann-Whitney test was utilized for non-normally distributed numerical data, while the Chi-Square or Fisher's Exact test was applied for categorical data. Statistical significance was set at a p-value < 0.05 with a 95% confidence interval.

Results

Initially, 35 participants met the inclusion criteria for this study. During the observation period, five subjects were lost to follow-up; therefore, the final data analysis was conducted on 30 subjects, equally divided into the SSW mineral nasal spray group (n=15) and the 0.9% NaCl saline group (n=15) (Table 1). Demographic characteristics showed a mean age of 36.4±14.09 years, with a female predominance of 21 participants (70%). Based on occupational profiles, housewives represented the largest group (36.7%), followed by employees (23.3%) and students (20%). Prior to the intervention, all subjects exhibited a high degree of inflammation, with 70% having a nasal mucosal eosinophil count in the +3 category and 30% in the +4 category. These clinical conditions were consistent with the baseline VAS scores, which indicated severe symptom intensity, particularly for the rhinorrhea parameter, with a median value of 9 (IQR 8–9) (Table 2).

Intra-group effectiveness on eosinophil cytology and VAS scores

Intra-group analysis demonstrated that both interventions resulted in statistically significant improvements ($p<0.001$) in the reduction of nasal mucosal eosinophil counts after four weeks (Table 3). In the 0.9% NaCl saline group, most subjects experienced

Table 1. Participant characteristics (n=30).

Variable	Value
Age (Years), Mean ± SD	36.4±14.09
Sex, n (%)	
Male	9 (30.0)
Female	21 (70.0)
Occupation, n (%)	
Housewives	11 (36.7)
Employees	7 (23.3)
Students	6 (20.0)
Entrepreneurs	4 (13.3)
Civil Servants	2 (6.7)
Treatment Group, n (%)	
Sea Salt Water Mineral Nasal Spray	15 (50.0)
0.9% NaCl Saline	15 (50.0)

Table 2. Distribution of baseline VAS scores and nasal mucosal eosinophil grades

Variable	Median (IQR) / n (%)
Baseline VAS Score	
Rhinorrhea	9 (8–9)
Nasal Obstruction	8 (7–9)
Recurrent Sneezing	8 (8–9)
Ocular Symptoms	8 (8–8)
Baseline Eosinophil Grade, n (%)	
+3	21 (70.0)
+4	9 (30.0)

a decrease from grade +3 to grade +1 (33.3%) and +2 (26.7%). More pronounced improvements were observed in the SSW mineral nasal spray group, where the majority of subjects with grade +3 at baseline improved to grade +1 (53.4%), and 20% of subjects achieved grade 0 (negative for eosinophils). Consistent with the cytological findings, VAS scores in both groups also showed significant progressive reductions ($p<0.001$) across all symptom parameters—including rhinorrhea, nasal obstruction, recurrent sneezing, and itchy/watery eyes—from the baseline period through the fourth week of observation.

Comparative analysis between treatment groups

The comparison of effectiveness between the two groups revealed interesting findings regarding the speed of symptom resolution. Based on the Mann-Whitney test, the SSW mineral nasal spray group showed a significantly faster reduction in VAS scores during the first two weeks compared to the 0.9% NaCl saline group ($p<0.001$). This was clearly observed in the parameters of rhinorrhea, recurrent sneezing, and ocular symptoms (Table 4). However, by the fourth week, the effectiveness of both solutions showed relatively similar final outcomes for nearly all symptoms, indicating that although isotonic saline works more slowly, its final clinical results approach the effectiveness of SSW. The total VAS score at the end of the study showed an equal median value for both groups, although the comparison of the change in total VAS score from 0–4 weeks still showed a statistically significant difference ($p=0.047$).

Table 3. Intra-group changes in nasal mucosal eosinophil counts following 4 weeks of therapy

Treatment Group	Pre-treatment, n (%)	Post-treatment, n (%)	p-value*
0.9% NaCl	+3: 10 (66.7)	0: 2 (13.4)	<0.001
	+4: 5 (33.3)	+1: 8 (53.3)	
		+2: 5 (33.3)	
Sea Salt Spray	+3: 11 (73.4)	0: 3 (20.0)	<0.001
	+4: 4 (26.6)	+1: 10 (66.6)	
		+2: 2 (13.4)	

Note: *Wilcoxon Test

Table 4. Comparison of VAS score reduction between treatment groups at various time intervals

VAS Score Change	Interval (Weeks)	0.9% NaCl, Median (IQR)	Sea Salt, Median (IQR)	p-value*
Rhinorrhea	0–2	-3 (-4 – -2)	-5 (-7 – -5)	<0.001
	0–4	-7 (-7 – -6)	-7 (-8 – -6)	0.440
Nasal Obstruction	0–2	-3 (-4 – -2)	-4 (-5 – -3)	0.036
	0–4	-6 (-8 – -6)	-5 (-6 – -4)	0.006
Recurrent Sneezing	0–2	-3 (-3 – -2)	-5 (-7 – -5)	<0.001
	0–4	-7 (-7 – -6)	-7 (-7 – -5)	0.216
Total VAS Score	0–2	-11 (-13 – -9)	-21 (-24 – -17)	<0.001
	0–4	-27 (-29 – -25)	-25 (-27 – -23)	0.047

Note: *Mann-Whitney Test

Discussion

This study provides clinical evidence regarding the efficacy of nasal irrigation as an adjuvant therapy in patients with moderate-to-severe persistent AR. The primary findings indicate that both 0.9% NaCl (isotonic saline) and SSW mineral nasal spray significantly reduce eosinophilic infiltration in the nasal mucosa and improve clinical symptoms as measured by VAS scores. However, a crucial point identified is the disparity in the speed of clinical response; the SSW group demonstrated significantly faster symptom recovery during the initial phase (week 2), suggesting that the trace mineral composition of seawater exerts a pharmacological role that extends beyond simple mechanical lavage.

Regarding demographic characteristics, the majority of subjects were female (70%) in their productive years. This is consistent with global literature indicating the influence of estrogen on increased nasal mucosal sensitivity and mast cell degranulation, explaining why the prevalence of AR is higher in adult females than in males (15–17). The high baseline VAS scores for rhinorrhea and nasal obstruction (median 8–9) across all subjects reflect a heavy disease burden. In moderate-to-severe persistent AR, a chronic inflammatory cycle occurs, leading to persistent epithelial damage; thus, interventions are required that are not only symptomatic but also capable of improving the nasal mucosal microenvironment.

The mechanism of action for 0.9% NaCl in this study can be explained through the theory of improved mucociliary clearance. Isotonic saline works by reducing mucus viscosity (a mucolytic effect) and flushing inflammatory mediators such as histamine, leukotrienes, and pro-inflammatory cytokines from the mucosal surface. Furthermore, the administration of 0.9% NaCl triggers the release of intracellular calcium and prostaglandin E2 (PGE2), which physiologically increases ciliary beat frequency (18). However, this process is gradual, as evidenced by the results where the reduction in VAS scores in the saline group tended to be linear but slower than that of the sea salt group.

The superiority of SSW mineral nasal spray during the initial phase (week 2) is closely related to its complex mineral content, including Magnesium (Mg^{2+}), Calcium (Ca^{2+}), and Potassium (K^{+}). Magnesium, specifically, acts as a natural calcium antagonist that inhibits calcium influx into mast cells, thereby preventing early degranulation and the release of chemical mediators. Additionally, these mineral elements play a role in maintaining the integrity of tight junctions in nasal epithelial cells, which are frequently damaged in AR patients (8). This anti-inflammatory and epithelial-protective effect is presumed to cause patients to experience more instant symptom relief (such as reduced sneezing and itching) compared to the use of standard saline alone. Beyond Magnesium (Mg^{2+}), Calcium (Ca^{2+}), and Potassium (K^{+}), SSW contains a rich matrix of Cl^{-}, Na^{+}, SO_4^{2-}, HCO_3^{-}, Br–, as well as essential trace elements such as zinc, selenium, iodine, iron, and manganese. This complex composition acts synergistically to moisturize the nasal mucosa, soften and facilitate the removal of crusts and thick exudate, and restore mucociliary clearance while enhancing local immunity. Crucially, the effective elimination of allergens and saturation of the mucosa with these beneficial trace elements through regular SSW irrigation significantly reduces the patient's reliance on pharmacological interventions, including nasal decongestants, systemic antihistamines, and topical corticosteroids. Furthermore, as seawater solutions are fully compatible with concurrent medications and possess a high safety profile, they are highly suitable for all patient demographics, including vulnerable groups such as pregnant and lactating women.

Cytologically, although both groups achieved equivalent results by week 4, the improvement in the sea salt group appeared qualitatively deeper, with 20% of subjects reaching an eosinophil score of 0 (negative). This reduction in eosinophils is critical, as eosinophils are primary effectors that release Major Basic Protein and Eosinophil Peroxidase, both of which are toxic to tissues. The similarity in final cytological outcomes at week 4 suggests a "plateau effect," where after a certain duration, the volume of lavage and the regularity of irrigation become more dominant factors than the solution's composition. This supports ARIA recommendations that position nasal irrigation as a primary supportive therapy that must be performed consistently (1,2).

The findings of this study offer practical guidance for the clinical management of moderate-to-severe persistent AR. While pharmacotherapy remains the standard of care, the integration of nasal irrigation provides a significant non-pharmacological advantage in reducing the local inflammatory burden. For clinicians, the disparity in symptom resolution speed between the two modalities suggests a tiered approach to treatment. SSW mineral nasal spray should be considered the preferred adjuvant for patients requiring rapid symptomatic relief to restore daily productivity and sleep quality during acute exacerbations. Conversely, 0.9% NaCl serves as a highly effective, accessible, and cost-efficient option for long-term maintenance and secondary prevention. By utilizing these irrigation strategies, clinicians may potentially reduce the patient's reliance on intranasal corticosteroids and antihistamines, thereby minimizing the risk of long-term pharmacological side effects.

A primary strength of this research lies in its dual-parameter evaluation, which correlates subjective patient-reported outcomes (VAS scores) with objective cellular inflammatory markers (nasal mucosal eosinophil cytology). Unlike many studies that rely solely on symptomatic assessment, the use of Papanicolaou-stained cytological scrapings provides a robust biological validation of the therapeutic effects of mineral-rich irrigation. The robust cytological findings in this study are largely attributed to the utilization of the Papanicolaou staining method. While conventionally associated with gynecology, this technique is highly informative for non-gynecological specimens, including nasal smears. Papanicolaou staining provides exceptional cytoplasmic transparency, allowing for clear visualization through cell layers, mucus, and decay products. This facilitates precise differentiation of eosinophils—the direct markers of AR—and allows for a detailed assessment of nuclear structures and epithelial conditions, including the presence of metaplasia, mast cells, goblet cells, and basophils. Furthermore, while previous literature occasionally questioned the value of nasal eosinophilia due to sampling limitations, our findings support that nasal smear examination is a highly specific, non-invasive, and cost-effective method. It effectively differentiates AR and non-allergic rhinitis with eosinophilia syndrome (NARES) from bacterial infections (characterized by neutrophilic infiltration) and serves as a valuable prognostic tool for predicting prolonged or recurrent eosinophilia. Furthermore, this study specifically targeted the moderate-to-severe persistent AR population, which represents the most clinically challenging group with the highest morbidity. The prospective design and strict inclusion criteria—requiring both ARIA-defined symptoms and positive skin prick tests—ensure that the observed improvements are directly attributable to the interventions. Lastly, this study contributes valuable regional data from the Indonesian population, specifically in Makassar, enhancing the diversity of global literature on nasal irrigation modalities.

The limitations of this study include the minimal sample size and reliance on self-irrigation techniques by the subjects, which may influence fluid distribution within the paranasal sinus cavities. Furthermore, the evaluation was only conducted up to week 4, meaning the long-term effect on recurrence has not yet been assessed. Additionally, this study focused exclusively on cytological inflammatory markers and subjective symptomatic parameters without evaluating dynamic functional characteristics, such as mucociliary clearance time or ciliary beat frequency. Future longitudinal studies incorporating these functional diagnostic assays are strongly recommended to broaden the evidence base and elucidate the complete mechanistic profile of mineral-enriched nasal irrigations. Nevertheless, clinically, the results of this study provide valuable guidance for clinicians: for patients requiring rapid symptom resolution to improve quality of life and productivity during the acute phase of AR, SSW mineral nasal spray offers a superior kinetic advantage. Meanwhile, 0.9% NaCl remains a highly effective and economical alternative for long-term maintenance.

Conclusion

Nasal irrigation using either 0.9% NaCl or SSW mineral nasal spray is significantly effective in reducing nasal mucosal eosinophil infiltration and improving VAS scores in patients with moderate-to-severe persistent AR. The primary finding indicates that SSW mineral nasal spray is superior in accelerating the resolution of clinical symptoms during the first two weeks of therapy. However, by the fourth week, both interventions demonstrated equivalent effectiveness across both subjective and objective parameters. This suggests that while the mineral content of seawater accelerates the initial response, irrigation consistency is the primary determinant of success in mid-term therapy.

Ethic Approval: All research designs were reviewed and approved by the Health Research Ethics Committee of Dr Wahidin Sudirohusodo Hospital, Faculty of Medicine, Hasanuddin University (No. 428/UN4.6.4.5.31/PP36/2025) on June 23, 2025.

Conflict of Interest: Each author declares that he or she has no commercial associations that might pose a conflict of interest in connection with the submitted article.

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Declaration on the use of AI: None

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